

Recitation: Rational Functions & Polynomial Long Division Solutions

Problem 1. Consider the rational function given below:

$$f(x) = \frac{3x^2 - 2x - 5}{(x+2)(x^8-1)} = \boxed{\frac{(3x-5)(x+1)}{(x+2)(x-1)(x+1)(x^2+1)(x^4+1)}} \quad , \quad = \quad , \quad \boxed{\frac{3x-5}{(x+2)(x-1)(x^2+1)(x^4+1)}}$$

Complete the following parts:

- (a) Find the domain.

The domain is where the original denominator is not 0. Solving $(x+2)(x-1)(x+1)(x^2+1)(x^4+1) = 0$, we see $x = -2, -1, 1$ (the x^2+1 and x^4+1 terms have no zeros). Therefore, the domain is $\mathbb{R} \setminus \{-2, -1, 1\}$.

- (b) Find any y -intercepts.

The y -intercept will be the value when $x = 0$. We have $f(0) = \frac{0-0-5}{2 \cdot -1} = \frac{-5}{-2} = \frac{5}{2}$. Therefore, the y -intercept is $\frac{5}{2}$.

- (c) Find any x -intercepts.

The x -intercepts are the solutions to $f(x) = 0$. For a rational function, this will be when the numerator is 0. Setting $(3x-5)(x+1) = 0$, we have $x = -1, \frac{5}{3}$. But the value $x = -1$ is not in the domain. Therefore, the only x -intercept is $\frac{5}{3}$.

- (d) Find any vertical asymptotes.

The vertical asymptotes will occur when the denominator is 0 in the 'reduced' rational function. So, we solve $(x+2)(x-1)(x^2+1)(x^4+1) = 0$, which implies $x = -2, 1$. Therefore, the vertical asymptotes are $x = -2$ and $x = 1$.

- (e) Find any horizontal asymptotes.

Examining the original rational function, the numerator has degree 2 and the denominator has degree 9. Therefore, the horizontal asymptote is $y = 0$.

- (f) Find any holes.

To find holes, we examine values not in the domain where the denominator is not 0 in the 'reduced' rational function. The only such value here is $x = -1$. Therefore, the only hole is at $x = -1$. The y -value will be the y -value at this x -value in the 'reduced' rational function. This is $\frac{-3-5}{1 \cdot -2 \cdot 2 \cdot 2} = \frac{-8}{-8} = 1$. Therefore, the hole is $(-1, 1)$.

- (g) Can this rational function be polynomial long divided? Explain.

No, the degree of the denominator is larger than the degree in the numerator. Therefore, this rational function cannot be polynomial long divided.

Problem 2. Complete the parts given below:

- (a) Factor the polynomial $x^3 - 25x$. Without explicitly computing, what is the remainder when this polynomial is divided by $x + 5$? Explain.
- (b) If $(3x^2 - x + 6)(x^2 + x - 3) + 2x - 4 = 3x^4 + 2x^3 - 4x^2 + 11x - 22$, then what is the polynomial quotient and remainder when $3x^4 + 2x^3 - 4x^2 + 11x - 22$ is divided by $x^2 + x - 3$? What about if it were instead divided by $3x^2 - x + 6$?
- (c) Without explicitly computing, can $x^2 - 4x + 6$ be divisible by $x^3 + 1$? Explain.
- (d) If a polynomial $f(x)$ is divisible by $x - c$, what is $f(c)$? Explain. Is the 'reverse' of your conclusion true? Explain.

Solution.

- (a) We have $x^3 - 25x = x(x^2 - 25) = x(x - 5)(x + 5)$. Observe that $x + 5$ is a factor of this polynomial. Therefore, the remainder upon division will be 0. We can also see that the quotient will be $x(x - 5) = x^2 - 5x$.
- (b) We can see that the quotient will be $3x^2 - x + 6$ and the remainder will be $2x - 4$. Similarly, when dividing by $3x^2 - x + 6$, the quotient will be $x^2 + x - 3$ and the remainder will also be $2x - 4$.
- (c) No. The degree of $x^2 - 4x + 6$ is 2 while degree of $x^3 + 1$ is 3, so we cannot long divide $x^2 - 4x + 6$ by $x^3 + 1$.
- (d) We know that a polynomial is divisible by $x - c$ if and only if $x = c$ is a root of $f(x)$. Therefore, because $x - c$ divides $f(x)$, we know $f(c) = 0$. The reverse is also true. Let's prove why.

First, let's see why $x = c$ is a root of $f(x)$ if $f(x)$ is divisible by $x - c$. Assume $f(x)$ is divisible by $x - c$. Polynomial long divide $f(x)$ by $x - c$. After this, we can express $f(x)$ as $f(x) = q(x)(x - c) + r(x)$, where $q(x)$ is the quotient polynomial and $r(x)$ is the remainder polynomial. But we assumed $x - c$ divided $f(x)$, so that $r(x) = 0$. But then $f(x) = q(x)(x - c)$. Observe that $f(c) = q(c)(c - c) = q(c) \cdot 0 = 0$, i.e. $x = c$ is a root of $f(x)$.

Now, let's see why $f(x)$ is divisible by $x - c$ if $x = c$ is a root of $f(x)$. Assume $x = c$ is a root of $f(x)$, i.e. $f(c) = 0$. Now polynomial long $f(x)$ by $x - c$. After this, we can express $f(x)$ as $f(x) = q(x)(x - c) + r(x)$, where $q(x)$ is the quotient polynomial and $r(x)$ is the remainder polynomial. Because $r(x)$ is the remainder polynomial, it has degree less than that of what we divided by— $x - c$. So, $r(x)$ must have degree 0, i.e. be a constant function. Suppose $r(x) = A$, where A is some number. We know $f(x) = q(x)(x - c) + r(x)$ for any value of x . Let's substitute $x = c$. We have $f(c) = q(c)(c - c) + r(c)$. But we know $f(c) = 0$ and the right side is $r(c)$. So, we have $0 = r(c)$. But $r(c) = A$, so $A = 0$, i.e. $r(x) = 0$. Therefore, $f(x) = q(x)(x - c) + r(x) = q(x)(x - c)$, i.e. $f(x)$ is divisible by $x - c$.

Problem 3. Polynomial long divide the expression below:

$$\begin{array}{r}
 \frac{5x^4 - 15x^3 - x^2 + 12x - 27}{x - 3} \\
 \\
 x - 3 \overline{) \begin{array}{r} 5x^3 \\ - 5x^4 + 15x^3 \\ \hline - x^2 + 12x - 27 \\ x^2 - 3x \\ \hline 9x - 27 \\ - 9x + 27 \\ \hline 0 \end{array}}
 \end{array}$$

Therefore, the quotient is $5x^3 - x + 9$ with remainder 0. So, we have $\frac{5x^4 - 15x^3 - x^2 + 12x - 27}{x - 3} = 5x^3 - x + 9$.

Problem 4. Polynomial long divide the expression below:

$$\begin{array}{r}
 \frac{2x^5 + 3x^4 - 6x^3 - 10x^2 - x + 9}{x^2 - 3} \\
 \\
 x^2 - 3 \overline{) \begin{array}{r} 2x^3 + 3x^2 - 1 \\ 2x^5 + 3x^4 - 6x^3 - 10x^2 - x + 9 \\ - 2x^5 + 6x^3 \\ \hline 3x^4 - 10x^2 - x + 9 \\ - 3x^4 + 9x^2 \\ \hline - x^2 - x + 9 \\ x^2 - 3 \\ \hline - x + 6 \end{array}}
 \end{array}$$

Therefore, the quotient is $2x^3 + 3x^2 - 1$ with remainder $-x + 6$. So, we have $\frac{2x^5 + 3x^4 - 6x^3 - 10x^2 - x + 9}{x^2 - 3} = 2x^3 + 3x^2 - 1 + \frac{-x + 6}{x^2 - 3}$.

Problem 5. Use the fact that $x = -1$ is a root of $9x^3 - 3x^2 - 5x - 1$ to fully factor this polynomial.

Solution. We know $x = a$ is a root of a polynomial if and only if $x - a$ is a factor of the polynomial. So, because $x = -1$ is a root, $x - (-1) = x + 1$ is a factor of $2x^3 + 3x^2 - 1$. Using polynomial long division, we have...

$$\begin{array}{r}
 \frac{2x^2 + x - 1}{x + 1} \\
 x + 1 \overline{) \begin{array}{r} 2x^3 + 3x^2 - 1 \\ - 2x^3 - 2x^2 \\ \hline x^2 \\ - x^2 - x \\ \hline - x - 1 \\ x + 1 \\ \hline 0 \end{array}}
 \end{array}$$

So, we know $2x^3 + 3x^2 - 1 = (2x^2 + x - 1)(x + 1) + 0 = (2x^2 + x - 1)(x + 1)$. The polynomial $2x^2 + x - 1$ factors as $(2x - 1)(x + 1)$. Therefore, $2x^3 + 3x^2 - 1 = (2x^2 + x - 1)(x + 1) = [(2x - 1)(x + 1)](x + 1) = (2x - 1)(x + 1)^2$.

$$\boxed{2x^3 + 3x^2 - 1 = (2x - 1)(x + 1)^2}$$